Summary in Graph

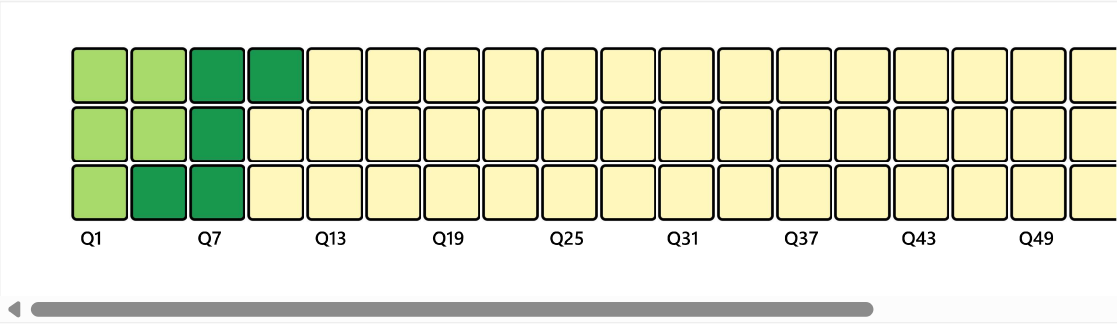
Exam Summary (GO Classes DA Test Series 2026 | MOCK GATE | Test 6)

Qs. Attempted:	**10** 5 + 5	**Correct Marks:**	**15** 5 + 10
Correct Attempts:	**10** 5 + 5	**Penalty Marks:**	**0** 0 + 0
Incorrect Attempts:	**0** 0 + 0	**Resultant Marks:**	**15** 5 + 10

Total Questions:	**65** 30 + 35
Total Marks:	**100** 30 + 70
Exam Duration:	**180 Minutes**
Time Taken:	**33 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (14) out of 135 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

If an integer is divisible by both 8 and 15, then the integer also must be divisible by which of the following?

A. 16
B. 24
C. 32
D. 36

Your Answer: B | Correct Answer: B | Correct | Time taken: 00min 49sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Pens cost 70 cents each and pencils cost 40 cents each. If Iris spent $\$5.20$ on a total of 10 pens and pencils, how many pencils did she purchase? ($\$1 = 100$ cents)

A. 4
B. 6
C. 8
D. 10

Your Answer: B | Correct Answer: B | Correct | Time taken: 02min 49sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

If 10^x equals 0.1 percent of 10^y, where x and y are integers, which of the following must be true?

A. $y = x + 2$
B. $y = x + 3$
C. $x = y + 3$
D. $y = 1,000x$

Your Answer: B | Correct Answer: B | Correct | Time taken: 00min 58sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

A large watermelon weighs 20 kg, with 98% of its weight being water. It is left to stand in the sun, and some of the water evaporates, so that now only 95% of its weight is water (i.e., 5% of its weight is non-water material). What is its reduced weight?

A. 17 kg
B. 19.4 kg
C. 10 kg
D. 8 kg

Your Answer: D | Correct Answer: D | Correct | Time taken: 07min 25sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Many diamond mines are _____ in South Africa.

A. Situation
B. Situate
C. Situated
D. Situating

Your Answer: C | Correct Answer: C | Correct | Time taken: 00min 13sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Consider the following definitions:

- A function is EVEN when $f(x) = f(-x)$
- A function is ODD when $f(-x) = -f(x)$

Which of the following statements are true?

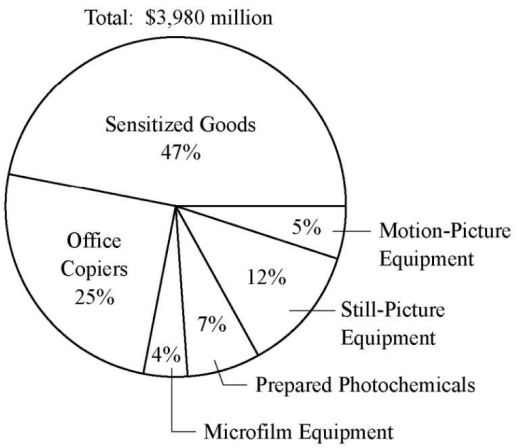
1. The product of 2 odd functions is always an odd function
2. The sum of 2 even functions is always an even function
3. The sum of 2 odd functions is always an odd function
4. The product of 2 even functions is always an even function

A. Only 1 is true
B. $2, 3, 4$ are true
C. All statements are true
D. None are true

Your Answer: B | Correct Answer: B | Correct | Time taken: 01min 33sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

A circle graph titled "United States Production of Photographic Equipment and Supplies in 1971" shows 6 categories, with total production at $\$3,980$ million.



If the value of office copiers produced in 1971 was 30% greater than in 1970, what was the approximate value of office copiers produced in 1970?

A. $\$765$ million
B. $\$995$ million
C. $\$1293$ million
D. $\$3980$ million

Your Answer: A | Correct Answer: A | Correct | Time taken: 05min 41sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

R is a list of 15 consecutive integers, and T is a list of 21 consecutive integers. The median of the integers in list R is equal to the least integer in list T. If the two lists are combined into one list of 36 integers, how many different integers are on the combined list?

A. 25
B. 27
C. 28
D. 32

Your Answer: C | Correct Answer: C | Correct | Time taken: 04min 44sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Andrew operates a vintage clothing business at a flea market, where he incurs a fixed daily rental fee of $\$150$ for a shop, plus a cost of $\$10$ per hour for an assistant. On average, he earns $\$78$ per hour in revenue from clothing sales. Assuming these are the only costs involved, which of the following equations best represents Andrew's profit for the day P in terms of the number of hours h that his shop is open?

A. $P(h) = 238 - 10h$
B. $P(h) = 72 - 10h$
C. $P(h) = 68h - 150$
D. $P(h) = 78h - 160$

Your Answer: C | Correct Answer: C | Correct | Time taken: 02min 54sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

Read the paragraph and choose the correct options.

Noodles were originally made using grains from millet, unlike the modern variety, which is made of wheat flour. Many cultures across the world claim to have invented it, but in 2005 the remains of the world's oldest noodles were unearthed in China. The yellow strands measuring 50 cm were found in a pot in Lajia along the Yellow River. It is believed that the pot got buried in an earthquake 4000 years ago.

We know that noodles were made thousands of years ago in China because the _____.

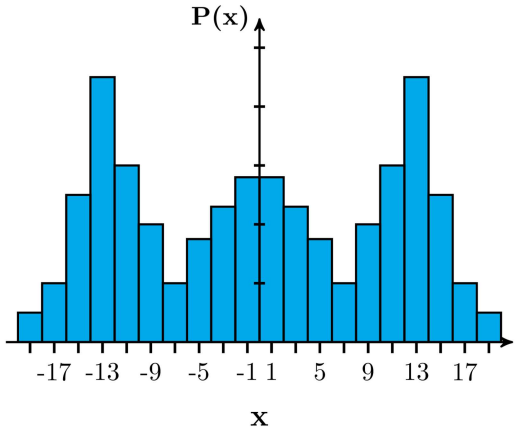
A. Chinese claim that they invented noodles.
B. Oldest noodles have been discovered in China.
C. Historians say so.
D. Chinese still eat noodles.

Your Answer: B | Correct Answer: B | Correct | Time taken: 03min 06sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Which one of the options can be inferred about the mean, median, and mode for the given probability distribution (i.e. probability mass function), $P(x)$, of a variable x?



A. mean = median ≠ mode
B. mean = median = mode
C. mean ≠ median = mode
D. mean ≠ mode = median

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

The probability density function of a continuous random variable distributed uniformly between x and y (for $y > x$) is

A. $\frac{1}{x-y}$
B. $\frac{1}{y-x}$
C. $x-y$
D. $y-x$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

The five items: A, B, C, D, and E are pushed in a stack, one after other starting from A. The stack is popped four items and each element is inserted in a queue. The two elements are deleted from the queue and pushed back on the stack. Now one item is popped from the stack. The popped item is

A. A
B. B
C. C
D. D

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in Python

Consider the following Python code:

```
    def f(n):
        print("A", end=" ")
        if n > 0:
            f(n-1)
5.      print("B", end=" ")

    print("C", end=" ")
    f(2)
```

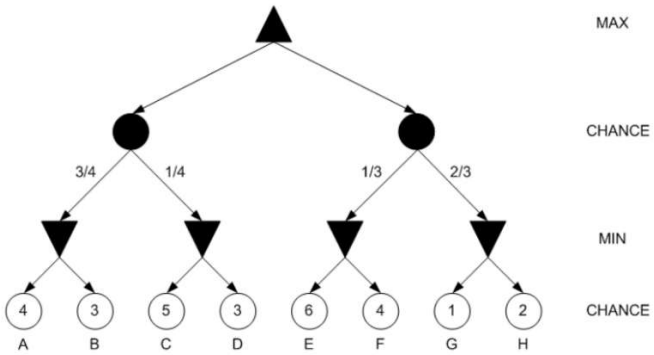
What will be the output when this code is executed?

A. C A B A B A B
B. C A A B A B A B
C. C A B A B
D. C A A A B B B

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Consider the following two-player game tree in which the game has an element of chance, which is shown by the so-called probability nodes in the game tree.



What is the correct value for the top MAX node if one applies the expectiminimax algorithm?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Given the dataset below, normalize the Income (₹) value of ₹$65,000$ using Min-Max normalization, where the new minimum value is 0 and the new maximum value is 1.

ID	Age (years)	Income (₹)	Credit Score
1	25	₹55,000	700
2	40	₹75,000	760
3	35	₹65,000	740
4	45	₹85,000	780
5	30	₹45,000	720

What is the normalized value of ₹$65,000$?

A. 0.25
B. 0.5
C. 0.75
D. 1

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 1 | Penalty: 0 | Artificial Intelligence

How many models does this propositional formula have:

$$(B \Rightarrow A) \vee A$$

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

We want to formalise the sentence "There exists a student who attends the Artificial Intelligence course but is not intelligent" in first-order logic. Given are the following predicates:

- $S(x)$: x is a student.
- $A(x, y)$: x attends course y.
- $I(x)$: x is intelligent.

We further introduce a term constant, AI, that stands for the Artificial Intelligence course. We are given two different formalisation

I. $\exists x S(x) \wedge A(x, AI) \Rightarrow \neg I(x)$
II. $\exists x (S(x) \wedge A(x, AI) \wedge \neg I(x))$

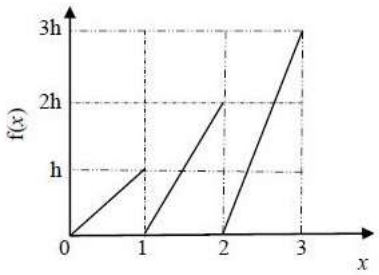
Which of the following assertions is correct?

A. Formalisation I is correct while formalisation II is incorrect.
B. Formalisation I is incorrect while formalisation II is correct.
C. Both formalisations are correct.
D. Both formalisations are incorrect.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

The graph of a function $f(x)$ is shown in the figure.



For $f(x)$ to be a valid probability density function, the value of h is

A. $1/3$
B. $2/3$
C. 1
D. 3

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Mark all statements that are true.

A. A B^{+} tree is always balanced-height
B. The keys on an internal node are always in sorted order
C. A leaf node can be a root node
D. Range queries are faster on B^{+} trees.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Numerical Type | Award: 1 | Penalty: 0 | Probability

If X and Y are random variables such that $E[2X+Y]=0$ and $E[X+2Y]=33$, then $E[X]+E[Y]=$ _____.

Your Answer: | Correct Answer: 11 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X be a real-valued random variable with $E[X]$ and $E[X^2]$ denoting the mean values of X and X^2, respectively. The relation which always holds true is

A. $(E[X])^2 > E[X^2]$
B. $E[X^2] \geq (E[X])^2$
C. $E[X^2] = (E[X])^2$
D. $E[X^2] > (E[X])^2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

In which of the below cases of hypothesis testing, a Type II error does occur?

A. When we reject a null hypothesis that is actually true in the population (Reject H_0, given that H_0 is true)
B. When we reject a null hypothesis that is actually false in the population (Reject H_0, given that H_0 is false)
C. When we fail to reject a null hypothesis that is actually true in the population (Accept H_0, given that H_0 is true)
D. When we fail to reject a null hypothesis that is actually true in the population (Accept H_0, given that H_0 is false)

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

If the matrix $A = \begin{bmatrix} a & 1 \\ 2 & 3 \end{bmatrix}$ has 1 as an eigenvalue, then $\textsf{trace}(A)$ is

A. 4
B. 5
C. 6
D. 7

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A be an $n \times n$ matrix. Consider the following statements.

i. A can have full-rank even if there exists two vectors $v_1 \neq v_2$ such that $Av_1 = Av_2$.
ii. A can be similar to the identity matrix, when A is not the identity matrix. Recall that two matrices B and C are said to be similar if $B = S^{-1}CS$ for some matrix S.
iii. If λ is an eigenvalue of A, then $\exists$ a vector $x \neq 0$ such that $(A - \lambda I)x = 0$.

Which of the above statements is/are TRUE? Choose from the following options.

A. Only (i)
B. Only (ii)
C. Only (iii)
D. (i), (ii), and (iii)

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider the matrix A and its reduced row echelon form below:

$$A = \begin{pmatrix} 1 & 1 & 2 & -2 \\ 1 & 0 & 1 & -2 \\ 2 & 1 & 3 & -4 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & 1 & -2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Which of the following is/are a basis for $\mathrm{Col}(A)$?

A. $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$

B. $\left\{ \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$

C. $\left\{ \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 2 \end{pmatrix} \right\}$

D. $\left\{ \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \right\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider a linear system $A\mathbf{x} = \mathbf{b}$, where A is a 4×5 matrix with $\text{Rank}(A) = 3$. The number of solutions to this system depends on the rank of the augmented matrix $[A \mid \mathbf{b}]$.

How many solutions does the system have in the following cases?

- **Case (a):** $\text{Rank}([A \mid \mathbf{b}]) = 3$.
- **Case (b):** $\text{Rank}([A \mid \mathbf{b}]) = 4$.

A.
Case (a): Infinitely many solutions.
Case (b): No solution.

B.
Case (a): No solution.
Case (b): Infinitely many solutions.

C.
Case (a): Unique solution.
Case (b): No solution.

D.
Case (a): Unique solution.
Case (b): Infinitely many solutions.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let the superscript T represent the transpose operation. Consider the function $f(x) = \frac{1}{2}x^T Q x - r^T x$, where x and r are $n \times 1$ vectors and Q is a symmetric $n \times n$ matrix. The stationary point of $f(x)$ is

A. $Q^T r$

B. $Q^{-1} r$

C. $\dfrac{r}{r^T r}$

D. r

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

The following dataset will be used to learn a decision tree for predicting whether a person is happy (H) or sad (S) based on the color of their shoes, whether they wear a wig, and the number of ears they have.

Color	Wig	Num. Ears	(Output) Emotion
G	Y	2	S
G	N	2	S
G	N	2	S
B	N	2	S
B	N	2	H
R	N	2	H

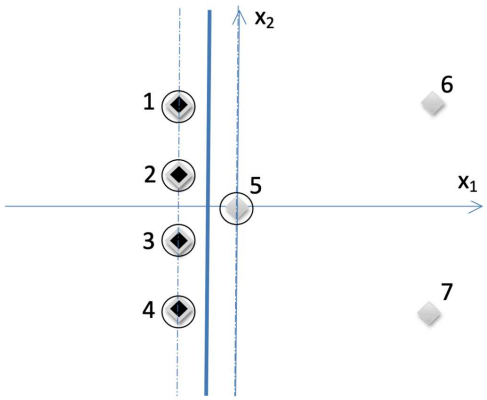
Color	Wig	Num. Ears	(Output) Emotion
R	N	2	H
R	N	2	H
R	Y	3	H

What is $\mathrm{H}(\text{ Emotion } \mid \text{ Wig } = \mathrm{Y})$?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider the following training points, where one class is distinguished from the other class points by their colors. The circled points are the support vectors.



What will be the leave-one-out cross-validation error for the SVM?

A. $1/7$
B. $2/7$
C. $4/7$
D. $5/7$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

In the usual formulation of soft-margin SVMs, each training sample has a slack variable $\xi_i \geq 0$ and we impose a regularization cost $C \sum_i \xi_i$. Consider an alternative formulation where we impose the additional constraints $\xi_i = \xi_j$ for all i, j. How does the minimum objective value $|\mathbf{w}|^2 + C \sum_i \xi_i$ obtained by the new method compare to the one obtained by the original soft-margin SVM?

A. They are always equal.
B. New minimum $\geq$ original SVM minimum.
C. Original SVM minimum $\geq$ new minimum.
D. New minimum is sometimes larger and sometimes smaller.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

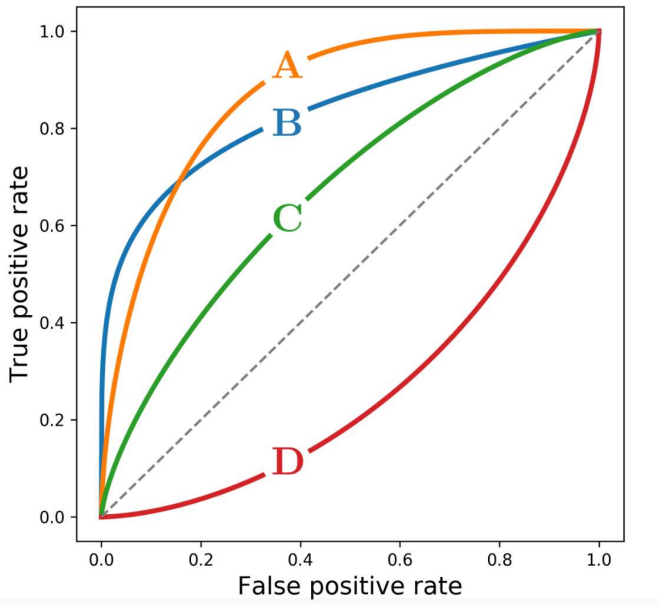
Which of the following quantities affect the bias-variance tradeoff?

A. λ, the regularization coefficient in ridge regression
B. ϵ, the learning rate in gradient descent
C. C, the slack parameter in soft-margin SVM
D. d, the polynomial degree in least-squares regression

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

You have trained four binary classifiers A, B, C, and D, observing the following ROC curves when evaluating them:



We say that a classifier G strictly dominates a classifier H if $G's$ true positive rate is always greater than $H's$ true positive rate for all possible false positive rates in $(0, 1)$.

Mark all of the below relations between $(A, B, C,$ and $D)$ which are true under this definition.

A. C strictly dominates D
B. D strictly dominates C
C. A strictly dominates B
D. B strictly dominates C

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

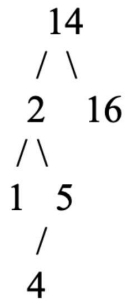
Consider the following dataset: $A = (0, 2), B = (0, 1)$ and $C = (1, 0)$. The k-means algorithm is initialized with centers at A and B. Upon convergence, the two centers will be at

A. A and C
B. A and the midpoint of BC
C. C and the midpoint of AB
D. A and B

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures

Consider the following binary search tree. Suppose we remove the root, replacing it with something from the left subtree. Which value will be the new root?



Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

With access to an n-by- n matrix of pairwise data distances, but no access to the data itself, we can use which of the following clustering techniques: A) k-means; B) k-medoids; C) hierarchical clustering:

A. Only A
B. Only A and B
C. Only B and C
D. A, B, and C

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Consider the following training dataset:

- Negative examples $(-1) : (1,1), (3,1), (1,4)$
- Positive examples $(+1) : (2,4), (3,3), (5,1)$

A test point is given at $(2,3)$. The perceptron algorithm is used with:

- Initial weights $w = (w0, w1, w2) = (-15, 5, 3)$, where $w0$ is the bias.
- Learning rate $= 0.5$

Simulate one iteration of the perceptron algorithm by processing the training points in the given order: $(1,1), (3,1), (1,4), (2,4), (3,3), (5,1)$.

Using the weights after one iteration, determine the predicted class of the test point $(2,3)$.

Your Answer: | Correct Answer: -1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

A classifier maps a feature vector x in a vector space X to a discrete label y. Two classifiers C_1 and C_2 are considered equal if, for all $x \in X, C_1(x) = C_2(x)$, i.e., the output labels produced by both classifiers are identical for every possible input in the feature space.

Suppose we have a training dataset S and two 1-Nearest Neighbor (1-NN) classifiers, C_1 and C_2, trained on S. The equality of the classifiers is analyzed in the following cases:

- **Case (a):** C_1 uses the L1-distance, and C_2 uses the L2-distance.
- **Case (b):** C_1 uses the L2-distance, and C_2 uses the square of the L2-distance.

Are C_1 and C_2 equal in the two cases described above?

A. C_1 is equal to C_2 in both case (a) and case (b).
B. C_1 is not equal to C_2 in case (a), but C_1 is equal to C_2 in case (b).
C. C_1 is equal to C_2 in case (a), but C_1 is not equal to C_2 in case (b).
D. C_1 is not equal to C_2 in either case (a) or case (b).

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider a Hard Margin Support Vector Machine (SVM) trained on only the two data points shown below. What are the values for α_i (the Lagrange multipliers) and the offset b that would give the maximal margin linear classifier for these points?

i	x_i	y_i
1	0	$+1$
2	4	-1

A. $\alpha_1 = \alpha_2 = \frac{1}{8}, b = 1$
B. $\alpha_1 = \alpha_2 = \frac{1}{4}, b = 0$
C. $\alpha_1 = 0, \alpha_2 = \frac{1}{2}, b = 1$
D. $\alpha_1 = \frac{1}{8}, \alpha_2 = \frac{1}{4}, b = -1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Given the training set below with boolean input features x, y, z and a boolean output U, what would be the predicted probability $P(U = 0 \mid x = 0, y = 1, z = 0)$ after training a Naive Bayes classifier?

x	y	z	U
1	0	0	0
0	1	1	0
0	0	1	0
1	0	0	1
0	0	1	1
0	1	0	1
1	1	0	1

A. $\frac{6}{35}$

B. $\frac{7}{35}$

C. $\frac{8}{35}$

D. $\frac{9}{35}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

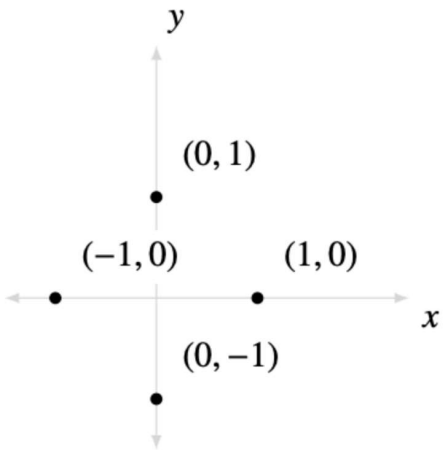
A classifier trained on $100,000$ training points with training accuracy of 99% gives accuracy of only 67% on testing data. Which of the following solutions can help us in improving the performance on the testing data?

A. We should gather more data for training.
B. We should set regularization parameter λ to 0.
C. We should reduce the number of training points.
D. We should use cross-validation to tune hyperparameters of the model.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 02min 19sec | Discuss

Q #32 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Consider running the **hierarchical agglomerative clustering algorithm** on the following set of four points in $\mathbb{R}^2$, breaking ties arbitrarily. If we stop when only two clusters remain, which of the following linkage methods ensures the resulting clusters are balanced (each have two sample points)? Select all that apply.



A. Complete linkage
B. Single linkage
C. Average linkage
D. None of the above

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider a recursive algorithm for sorting an array of $n \geq 2$ integers that works as follows.

a. If there are only 2 elements to be sorted, compare them and swap them if they are out of order.
b. Otherwise, do the following steps in order.
    1. Recursively sort the first $n-1$ elements of the array.
    2. In the resulting array, recursively sort the last $n-1$ elements.
    3. In the resulting array, recursively sort the first 2 elements of the array.

What is the asymptotic running time complexity of this algorithm measured in terms of the number of comparisons made?

A. $\Theta(n \log n)$
B. $\Theta\left(n^2\right)$
C. $\Theta\left(n^3\right)$
D. $\Theta\left(2^n\right)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Let $T(n) = 2T(n/2) + O(n)$, where "O" is big-oh. What will be asymptotic bound on $T(n)$?

A. $\Theta(n)$
B. $\Theta(n \log n)$
C. $\Theta\left(n^2\right)$
D. $O\left(n^3\right)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

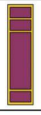
Q #35 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider the following unsorted array, and the same array after 4 iterations of insertion sort (where we sweep through the array from left to right, and move an element to its relative correct position).

Assume no two elements are equal. Insertion sort arranges the elements from least to greatest.



Which of the following is necessarily the correct relation <, or > between objects?

	Symbol 1		Symbol 2
A.		<	
B.		>	
C.		<	
D.		>	

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider the implementation of a stack of which is based on dynamic array allocation. This implementation allows us to push more elements than the current array capacity. It works as follows.

Suppose the array is full, then to push the next element we first allocate a new array of larger size, copy all elements to the new array, and discard the old array.

When a dynamic array-based stack becomes full, you increase the array size by the following sequence $k, 2k, 3k, 4k, \ldots$ for some positive constant k. Which means the first time we increase the size by k, next time by $2k$, and so on.

If you have an empty stack that uses an array whose initial size is k, and you perform n pushes (assume that $n > k$). What is the total cost complexity of executing n push operations?
Let cost of expanding from k to $2k$ has $2k$ cost, similarly cost of expanding from $2k$ to $3k$ has $3k$ cost.

A. $O\left(n^2/k\right)$
B. $O(n/k)$
C. $O\left(n^2k\right)$
D. $O(nk)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

We consider log to the base of 2. Thus for $\log \mathrm{N}$ to be a natural number, N must be a power of 2. Let N be an arbitrary number with $\mathrm{N} > 1$. Which one of the following statements is true? (Assume root is at height zero)

A. A binary tree of height $\log \mathrm{N}$ can have exactly N nodes.
B. A binary tree of height $\log \mathrm{N}$ can have exactly $\log \mathrm{N}$ nodes.
C. A binary tree of height $\log \mathrm{N}$ can have exactly $\mathrm{N} \log \mathrm{N}$ nodes.
D. A binary tree of height $\log \mathrm{N}$ can have exactly N^2 nodes.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in Python

What will be the output of the following Python code?

```
    def mystic(lst):
        if len(lst) <= 1:
            return lst
        return mystic(lst[1:]) + [lst[0]]
5.
    my_list = [1, 2, 3, 4, 5]
    result = mystic(my_list)
    print(result)
```

A. $[5, 4, 3, 2, 1]$
B. $[1, 2, 3, 4, 5]$
C. $[5, 4, 3]$
D. $[\,]$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Underlying each search problem is a *search graph* in which the states are the vertices of the search graph and the edges (connections) are determined by the possible actions; there is an edge from s to s' with label a in the search graph, if and only if there is an action a which leads from state s to state s'. These two conditions together completely define the search graph. A cycle in the *search graph* is a vertex (state) s and a non-empty sequence of actions as such that if we start in s and execute the sequence of actions as then we will end up in s again. Consider the following statements about search problems and graphs:

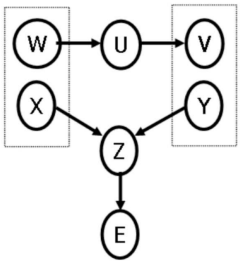
i. If the **Breadth-First Search Graph Algorithm** does terminate on a given search problem then the corresponding search graph contains no cycles.
ii. If the corresponding search graph is finite and contains no cycles then the **Depth First Search Tree Algorithm** always terminates on the corresponding search problem.

Which of the following claims is true?

A. Both statements (i) and (ii) are false.
B. Only statement (i) is true.
C. Both statements (i) and (ii) are true.
D. Only statement (ii) is true.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence



Consider the Bayesian network above.

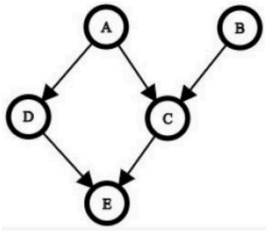
Let S be a set of variables such that $\{\mathrm{W}, \mathrm{X}\}$ is d-separated from $\{\mathrm{V}, \mathrm{Y}\}$ given S . Which of the following statements is/are true?

A. Variables U, Z and E must be in set S
B. Variables U and E must NOT be in set S
C. Variables U and Z must be in set S
D. Variables Z and E must NOT be in set S

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Consider the following Bayesian Network.



Statement 1 : B is independent of E given evidence at C.

Statement 2 : B is independent of A given evidence at E.

A. Both statements are correct.
B. Both statements are incorrect.
C. Statement 1 is correct and Statement 2 is incorrect.
D. Statement 1 is incorrect and Statement 1 is correct.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following schema for bike riders:

```
CREATE TABLE Riders (rid INTEGER PRIMARY KEY, name VARCHAR);
CREATE TABLE Bikes (bid INTEGER PRIMARY KEY, owner INTEGER REFERENCES riders(rid)););
CREATE TABLE Rides (rider INTEGER REFERENCES riders(rid), bike INTEGER REFERENCES bikes(bid));
```

Select all of the following queries which return the bid of all Bikes that have never been ridden.

```
A. SELECT bid FROM bikes b1 WHERE NOT EXISTS
        (SELECT owner FROM bikes b2 WHERE b2.bid = b1.bid);

B. SELECT bid FROM bikes WHERE NOT EXISTS
        (SELECT bike FROM rides WHERE bike = bid);

C. SELECT bid FROM bikes WHERE bid NOT IN
        (SELECT bike FROM rides, bikes as b2 WHERE bike = b2.bid);
```

Which queries are correct?

A. Only B
B. Only C
C. Only B, C
D. All

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider relation $R = ABCDEF$ and functional dependencies $F = \{AB \rightarrow CF, CE \rightarrow B, F \rightarrow D\}$.

Consider the following statements:

- A : The decomposition of R into $ABCE$ and $ABDEF$ is a dependency preserving decomposition.
- B : The decomposition of R into $ABCE$ and $ABDEF$ is a lossless decomposition.

Which of the above statements are correct?

A. Only A
B. Only B
C. Both A &B
D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Probability density function of a random variable X is given below

$$f(x) = \begin{cases} 0.25 & \text{if } 1 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

$P(X \leq 4)$ is

A. $\frac{3}{4}$

B. $\frac{1}{2}$

C. $\frac{1}{4}$

D. $\frac{1}{8}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Numerical Type | Award: 2 | Penalty: 0 | Probability

Consider the following probability mass function (p.m.f) of a random variable X :

$$p(X = x) = \begin{cases} q & \text{if } x = 0 \\ 1-q & \text{if } x = 1 \\ 0 & \text{otherwise} \end{cases}$$

If $q = 0.4$, the variance of X is _____

Your Answer: | Correct Answer: 0.23 : 0.25 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let Z be an exponential random variable with mean 1. The cumulative distribution function of Z is given by

$$F_Z(x) = \begin{cases} 1 - e^{-x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

What is $\Pr(Z > 2 \mid Z > 1)$?

A. e^{-2}
B. e^{-1}
C. e^{2}
D. e^{1}

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Two random variables X and Y are distributed according to the following joint probability density function (PDF):

$$f_{X,Y}(x,y) = \begin{cases} x + y, & 0 \leq x \leq 1, \quad 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

What is the probability $P(X + Y \leq 1)$?

A. $\frac{1}{2}$

B. $\frac{1}{3}$

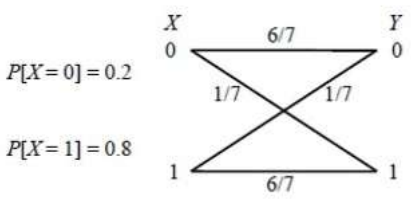
C. $\frac{1}{4}$

D. $\frac{1}{6}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Numerical Type | Award: 2 | Penalty: 0 | Probability

The input X to the Binary Symmetric Channel (BSC) shown in the figure is $'1'$ with probability 0.8. The cross-over probability is $1/7$. If the received bit $Y = 0$, the conditional probability that $'1'$ was transmitted is



Your Answer: | Correct Answer: 0.39 : 0.41 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Suppose m and n are positive integers, $m \neq n$, and A is an $m \times n$ matrix with real entries. Which of the following statements is/are true for all such A?

A. $\text{rank}(AA^T) = \text{rank}(A^TA)$
B. $\det(A^TA) = \det(AA^T)$
C. Trace $(AA^T) = \text{Trace}(A^TA)$
D. The nonzero eigenvalues of AA^T and A^TA are the same.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

The maximum value of the determinant among all 2×2 real symmetric matrices with trace 14 is _____.

Your Answer: | Correct Answer: 48.9 : 49.1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let M be a real 4×4 matrix. Consider the following statements:

$S1$: M has 4 linearly independent eigenvectors.

$S2$: M has 4 distinct eigenvalues.

$S3$: M is non-singular (invertible).

Which one among the following is TRUE?

A. $S1$ implies $S2$
B. $S1$ implies $S3$
C. $S2$ implies $S1$
D. $S3$ implies $S2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following statements about vector spaces and linear transformations is/are correct?

A. If $v \in \text{Span}\{w, a\}$ and $w \in \text{Span}\{a, b\}$, then $v \in \text{Span}\{a, b\}$, where v, w, a, b are vectors in a vector space.

B. If V and W are n-dimensional vector spaces, then there exists a linear transformation $T : V \to W$ that is both $1-1$ and onto.

C. If V is an n-dimensional vector space, then every linearly independent subset of V contains at most n vectors.

D. If $T : \mathbb{R}^n \to \mathbb{R}^n$ is a linear transformation, $[T]$ is the standard matrix of T, and $[T]_{\mathcal{B}}$ is the matrix of T with respect to another basis $\mathcal{B}$, then $\det([T]) = \det([T]_{\mathcal{B}})$.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the following two statements regarding linear transformations:

- **S1:** If $v_1, \ldots, v_n$ are linearly independent vectors in V, and T is a linear transformation from V to V, then $T(v_1), \ldots, T(v_n)$ are linearly independent.
- **S2:** If $T(v_1), \ldots, T(v_n)$ are linearly independent vectors in V, where T is a linear transformation from V to V, then $v_1, \ldots, v_n$ are linearly independent.

Which of the following is true?

A. Only S1 is true.
B. Only S2 is true.
C. Both S1 and S2 are true.
D. Neither S1 nor S2 is true.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Select all choices that are subspaces of $\mathbb{R}^3$.

Note: $\mathbb{R}$ denotes the set of real numbers.

A. $W_1 = \left\{(a_1, a_2, a_3) \in \mathbb{R}^3 \mid a_1 = 3a_2 \text{ and } a_3 = -a_2\right\}$
B. $W_2 = \left\{(a_1, a_2, a_3) \in \mathbb{R}^3 \mid a_1 = a_3 + 2\right\}$
C. $W_3 = \left\{(a_1, a_2, a_3) \in \mathbb{R}^3 \mid a_1 < a_2\right\}$
D. $W_4 = \left\{(a_1, a_2, a_3) \in \mathbb{R}^3 \mid 5a_1^2 - 3a_2^2 - 6a_3^2 = 0\right\}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following functional dependencies in a database.

$$\mathrm{M} \to \mathrm{O}, \mathrm{NO} \to \mathrm{P}, \mathrm{L} \to \mathrm{MN}$$

The relation $\mathrm{R}(\mathrm{L}, \mathrm{M}, \mathrm{N}, \mathrm{P})$ is

A. in second normal form but not in third normal form
B. in third normal form but not in BCNF
C. in BCNF
D. in none of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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